ACSL Contest 3 Quiz: Stacks/Queues and Boolean Algebra

Total: 16 Points

12 minutes

* Required

Student Information

1

Name: *

2

Grade *

Number must be between 6 ~ 8

3

Student Email (ending in @lwsd.org) *

Please enter an email

Stacks and Queues

4

A stack currently contains, from bottom to top: 3, 5, 7. Which sequence of operations results in 5 being on top? * (1 Point)

- POP, POP, PUSH(5)
- PUSH(2), POP(X)
- POP(X), PUSH(7)
- PUSH(5), POP(X)

5

Which of the following is **NOT true** about queues? * (1 Point)

- Elements leave in the order they arrive
- Can be implemented using arrays or linked lists
- Always follows LIFO
- Useful for scheduling tasks

6

Given the stack operations: PUSH(1), PUSH(2), PUSH(3), POP(X), PUSH(4), what is the top element? * (1 Point)

The value must be a number

7

Given the stack operations: PUSH(1), PUSH(2), PUSH(3), POP(X), PUSH(4), what elements are in the stack from bottom to top? *(separate items with commas)* * (1 Point)

8

A queue initially contains 2, 4, 6 (front to back). Perform the operations: POP(X), PUSH(8), POP(X). What is the queue what elements are in the stack from front to back? *(separate items with commas)* * (1 Point)

9

A stack contains 10, 20, 30 (front to back). Which sequence of operations leaves 20 at the front of the stack? * (1 Point)

- POP(X), PUSH(20)
- POP(X), PUSH(10)
- POP(X), PUSH(40), POP(X)
- PUSH(40), POP(X)
- POP(X)

Boolean Algebra

10

Simplify this expression: *
(1 Point)

$$A \cdot (A + B)$$

- 0
- A + B
- A
- B

11

Which expression is equivalent to:
* (1 Point)

$$\overline{A + B} \cdot C$$

$$\overline{A} + \overline{B} +$$

Option 1

$$(\overline{A} \cdot \overline{B}) \cdot$$

Option 2

$$\overline{A \cdot B} \cdot C$$

Option 3

$$A \cdot B +$$

Option 4

12

Simplify this:
* (1 Point)

$$(A + \overline{B}) \cdot (A + B)$$

- 0
- A
- B
- A + B

13

Evaluate if A=1, B=0, C=1: * (1 Point)

$$(A + \overline{B}) \cdot (B + C)$$

- Cannot Determine
- 0
- 1
- 2

14

True or False: * (1 Point)

$$A + \overline{A} \cdot B = A + B$$

- True
- False

15

Simplify: * (1 Point)

$$\overline{(A \cdot B)} + A$$

- A + B
- 1
- $\overline{A} + \overline{B}$
- $\overline{B} + A$
- Option 3
- Option 4

16

Simplify: * (1 Point)

$$(A+B)\cdot(A+\overline{B})\cdot(A+C)$$

- A + B + C
- A + B
- A
- B + C

17

Evaluate this expression for X = 1, Y = 0: * (1 Point)

$$\overline{X}\cdot Y + X\cdot\overline{Y}$$

- 0
- 1
- 2
- Cannot determine

18

Which Boolean expression corresponds to XOR of A and B? *
(2 Points)

$$A \cdot B$$

Option 1

$$A + B$$

Option 2

$$\overline{A} \cdot B + A \cdot$$

Option 3

$$\overline{A + B}$$

Option 4

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